

Atishaya Maharjan

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EDUCATION

BSc. Joint Honours: Computer Science & Mathematics, Minor in Statistics Aug. 2022 – Present
University of Manitoba *Winnipeg, Canada*

International Undergraduate Student Scholarship | Dean's Honour Roll

PROFESSIONAL EXPERIENCE

Software Engineer Mar. 2024 – Present
LG Electronics *Englewood Cliffs, USA (Remote)*

- Shipped **Next.js/Redux/TypeScript** B2B platform features for **1,000+** vendors, contributing to a **25%** user-engagement lift.
- Designed analytics and tagging frameworks adopted by **6+** product teams, enabling real-time measurement and data-driven feature prioritization.
- Built AI-powered global search with **35 ms** lower latency and checkout workflows that drove a **15%** conversion lift across promo-code and magic-link flows.

Software Engineer | Vibecore – *Cloud Infrastructure & CI/CD*

- Automated CI/CD deployments from **GitHub Actions** to **AWS EC2**, standardizing releases for EC2-hosted applications.
- Configured **S3**, **DynamoDB**, and **IAM** permissions for cloud-connected application workflows.
- Provisioned and maintained **EBS/EFS** storage for application data and deployment infrastructure.

Software Engineer Jun. 2025 – Jan. 2026
Hoek Insights *Amsterdam, Netherlands (Remote)*

- Engineered a **Python** package for **Three.js** visualizations in **Jupyter**, reducing render time by **38%** and bundle size by **25%** through **React** renderer optimization.
- Shipped a **FastAPI/PostgreSQL** MVP with **18+** endpoints and **10** workflows; maintained below **200 ms** median staging latency and **95%** integration-test coverage.

Software Engineer Aug. 2019 – Apr. 2024
Timeero *Phoenix, USA (Remote)*

- Led frontend development of a **TypeScript/React/Vite/Mapbox** mileage-tracking application supporting **1,000+** active users.
- Shipped **15+** geospatial workflows for segment mapping and mileage capture, improving time-capture accuracy by **30%**.

RESEARCH EXPERIENCE

Undergraduate Researcher – Algorithms & Computational Geometry Mar. 2024 – Present
Geometric, Approximation, and Distributed Algorithms (GADA) Lab *Winnipeg, Canada*

- Optimizing collaborative drone-delivery routes for logistics by developing exact and approximation algorithms with dynamic programming, exchange-point geometry, and structural analysis.
- Developed an approximation algorithm for Hamiltonian-path optimization in spatially embedded graphs [↗](#), together with visualization and randomized testing infrastructure covering **100+** synthetic instances.
- Improved the performance of a noncrossing spanning-tree research library by **30%** and introduced stronger **Python** testing, documentation, and development practices.

Machine Learning Researcher May. 2025 – Present
Terrabyte Machine Learning Research Group *Winnipeg, Canada*

- Extended diffusion-based learning to higher-order relational data by implementing **4** Hypergraph Neural Diffusion variants with **PyTorch** and **PyTorch Geometric**.
- Built a reproducible **MLflow** pipeline tracking configurations, metrics, and evaluations for **25+** experiments across **5+** datasets.

Undergraduate Researcher – Combinatorics & Graph Theory

Jan. 2024 – Jul. 2025

Department of Mathematics, University of Manitoba

Winnipeg, Canada

- Established a coefficientwise inequality underlying the Hurlbert–Kamat property for perfect m -ary trees  and derived a closed-form expression for their independence number.
- Applied generating functions, recurrence analysis, induction, and **SageMath** experiments across **1,000+** instances to develop the proof, test conjectures, and verify symbolic identities.

CS Help Center Leader & Teaching Assistant

Jan. 2025 – Apr. 2025

University of Manitoba

Winnipeg, Canada

- Mentored **100+** students in first- through third-year CS courses and graded **150+** assignments for Analysis of Algorithms (COMP 3170).

SELECTED PROJECTS

PayTime | *Next.js, FastAPI, Supabase*

1st Place

- Won **1st place** at the CSSA Design & Craft Case Competition 2026 by building an auditable payroll-ingestion platform in a **6.5-hour** technical sprint.
- Implemented field mapping, validation, exception review, and source traceability for **CSV, XLSX, PDF**, image, and email-derived records using **Next.js, TypeScript, FastAPI, Python**, and **Supabase**.

HAPA | *Next.js, TypeScript, AI/LLM*

Winner

- Won the **AccelR8 Prairie AI Hackathon 2026** with a working shopping prototype that learns preferences and dynamically personalizes product feeds.
- Combined onboarding signals, swipe feedback, and natural-language/voice pivots in a **12-hour** sprint, with an extensible architecture for multimodal personalization and AI-assisted commerce.

WinLocate | *Full-Stack, Open Data, AI*

1st Place

- Built a **Socrata**-powered relocation platform with city comparisons, interactive zoning maps, budget calculators, and decision-support workflows for companies considering Winnipeg.
- Won **1st place** among **15** teams at the Winnipeg Open Data Datathon 2026.

SELECTED HONOURS & AWARDS

S.U.R.E. Poster Competition – 3rd Place, Computational Science

Aug. 2024

- Recognized for the research poster “*Minimizing Maximum Interference in 1-Dimensional Networks*” at the University of Manitoba.

CSSA Design & Craft Case Competition – 1st Place

2026

AccelR8 Prairie AI Hackathon – Winner

2026

Winnipeg Open Data Datathon – 1st Place among 15 teams

2026

CSSA Capture the Flag – 1st of 41 teams (2026), 2nd of 25+ teams (2025)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C, C++, Java, R, SQL, Bash, SageMath

Frameworks & Libraries: React, Next.js, Redux, FastAPI, Node.js, Astro, Electron, Three.js

Data & Machine Learning: PostgreSQL, Redis, MongoDB, DynamoDB, Pinecone, PyTorch, PyTorch Geometric, MLflow

Cloud & Infrastructure: AWS EC2, S3, DynamoDB, IAM, EBS/EFS, Docker, GitHub Actions, Linux

Engineering Practices: CI/CD, API design, integration testing, automated testing, reproducible experimentation

Specializations: Full-stack development, cloud infrastructure, ML systems, geospatial systems

COMMUNITY & MENTORSHIP

Coding Kickoff Facilitator: taught Git and IDE setup to **50+** students; **.devClub Events Executive:** co-led technical workshops, hackathons, and academic review sessions.